

Low-Dimensional 2-thiopheneethylammonium Lead Halide Capping Layer Enables Efficient Single-Junction Methylamine-Free Wide-Bandgap and Tandem Perovskite Solar Cells

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Wide-bandgap (WBG) perovskite solar cells (PSCs) have garnered significant attention for their potential applications in tandem solar cells. However, their large open-circuit voltage (V_{OC}) deficit and serious photo-induced halide segregation remain the main challenges that impede their applications. Herein, a post-treatment strategy without thermal annealing is presented to form a 2D top layer of 2-thiopheneethylammonium lead halide ($n = 1$) on WBG perovskites. This thermal annealing-free post-treatment method can more effectively passivate the defects of WBG methylamine (MA)-free formamidinium/cesium lead iodide/bromide perovskite films and suppress photo-induced perovskite phase segregation, as compared with the thermal annealing method that yields multi-2D phases. The resulting opaque and semi-transparent 1.66 eV-bandgap perovskite solar cells deliver maximum power conversion efficiencies of 21.47% (a small V_{OC} deficit of 0.43 V) and 19.11%, respectively, both of which are among the highest reports for inverted MA-free WBG PSCs. Consequently, four-terminal all-perovskite tandem cells realize a remarkable efficiency of 26.64%, showing great promise for their applications in efficient multi-junction tandem solar cells.

narrow-bandgap perovskite,^[4–9] perovskite/crystalline silicon,^[10–14] perovskite/copper indium gallium diselenide,^[15–18] and perovskite/organic tandem cells.^[19–21] Among them, WBG PSCs are a critical component in achieving highly efficient perovskite-based TSCs. However, WBG PSCs are suffering from severe open-circuit voltage (V_{OC}) deficit (defined as $E_g/e \cdot V_{OC}$, where E_g is the bandgap of perovskite thin films), which is mainly attributed to high trap-state densities at grain boundaries and surface-produced non-radiative recombination pathways.^[22,23] In addition, photo-induced perovskite phase segregation leads to bandgap instabilities.^[24,25] These inferior characteristics seriously diminish their photovoltaic performance.^[26,27]

To date, various strategies have been proposed to improve the quality of WBG perovskite films,^[28–32] which can achieve a large V_{OC} of PSCs. It has been reported that the introduction of addi-

tives is a good choice to enhance the crystal quality of perovskite films.^[33–37] For example, Zheng et al. recently added 4-(2-aminoethyl)-benzenesulfonyl fluoride hydrochloride into mixed-halide formamidinium/methylamine lead iodide/bromide ($\text{FA}_{0.75}\text{MA}_{0.25}\text{Pb}(\text{I}_{0.75}\text{Br}_{0.25})_3$) perovskite precursors for suppressing the initial vertical halide phase separation, which finally gained a PCE of 20.1% and a greatly reduced V_{OC} loss of 0.47 V.^[35] Tao et al. employed F-type pseudo-halogen additives (PF_6^- or BF_4^-) into $\text{FA}_{0.65}\text{MA}_{0.20}\text{Cs}_{0.15}\text{Pb}(\text{I}_{0.8}\text{Br}_{0.2})_3$ perovskite precursors, which exhibited a high PCE of 20.6% with a V_{OC} deficit of 0.47 V and effectively inhibited the phase separation.^[36] Ye et al. introduced complementary additives based on pentafluorophenethylammonium iodine (F_5PEAI) and lead thiocyanate ($\text{Pb}(\text{SCN})_2$) to improve the quality of $\text{FA}_{0.64}\text{MA}_{0.20}\text{Cs}_{0.15}\text{Pb}_{0.99}(\text{I}_{0.79}\text{Br}_{0.2})_3$ thin films and enable 1.68 eV-bandgap PSCs with PCEs over 21%.^[37] In addition, interface defect passivation is also a widely used tactic to reduce non-radiative recombination pathways in PSCs.^[38–41] For instance, Liu et al. made use of an MgF_x interlayer with a thickness of ≈ 1 nm at the $\text{Cs}_{0.05}\text{FA}_{0.8}\text{MA}_{0.15}\text{Pb}(\text{I}_{0.75}\text{Br}_{0.25})_3$ /electron transport layer (i.e., C_{60}) interface and exhibited a low V_{OC} deficit of 0.46 V.^[39] Azmi et al. adopted an oleylammonium iodide molecule to tune the surface work function of $\text{Cs}_{0.05}\text{FA}_{0.80}\text{MA}_{0.15}\text{PbI}_{2.25}\text{Br}_{0.75}$

1. Introduction

The power conversion efficiency (PCE) of perovskite solar cells (PSCs) has enhanced remarkably in the past decades, rising from 3.8%^[1] in 2009 to 25.7%^[2] in 2023, which is close to attaining their Shockley–Queisser (S–Q) limits of single-junction solar cells.^[3] To further improve the PCE, multi-junction solar cells hold the potential to raise the PCE beyond the S–Q limits. Specifically, perovskite-based tandem solar cells (TSCs) composed of a top wide-bandgap (WBG) perovskite subcell and a bottom narrow-bandgap (NBG) subcell have been considered as promising alternatives, such as wide-bandgap perovskite/

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films, which exhibited a high PCE of 20.8%.^[40] While, as the mentioned approaches above, most highly efficient WBG PSCs have to incorporate volatile MA cation to attain high PCEs. However, the incorporation of MA into WBG perovskites is a double-edged sword, which can increase solar cell efficiency but decrease device stability because MA cations have poor thermal stability,^[42] hindering their future commercialization. Therefore, it is very crucial to develop a method that simultaneously achieves low V_{OC} deficit and high PCE for MA-free WBG PSCs.

Herein, we report a strategy to reduce the V_{OC} deficit and achieve highly-efficient MA-free WBG PSCs. Our approach involved a post-treatment method that did not require thermal annealing, which formed a 2D perovskite capping layer using 2-thiopheneethylammonium chloride (TEACl). Compared to similar methods that employed thermal annealing and other reports that introduced chaotic multi-2D phases,^[30,38,40] this thermal annealing-free post-treatment method gave rise to a primary 2D ($n = 1$) member of 2-thiopheneethylammonium lead halide (TEA_2PbX_4 , $X = Cl, Br, I$) top layer on WBG perovskites. The resulting WBG perovskite films had numerous advantages, such as no use of MA cation, a significant reduction in defects, a much-prolonged carrier lifetime, and a significant suppression of photo-induced perovskite phase segregation. As a result, the as-prepared opaque MA-free $FA_{0.75}Cs_{0.25}Pb(I_{0.8}Br_{0.2})_3$ WBG PSCs ($E_g \approx 1.66$ eV) presented a remarkable PCE of 21.47% with a low V_{OC} deficit of 0.43 V and a high fill factor (FF) of 84.33%. In addition, the semi-transparent WBG devices achieved an impressive PCE of 19.11%, enabling four-terminal (4T) perovskite/perovskite TSCs with a promising PCE of 26.64% by coupling 1.25 eV-bandgap mixed Pb-Sn perovskite bottom cells.

2. Results and Discussion

We assembled inverted WBG PSCs using MA-free $FA_{0.75}Cs_{0.25}Pb(I_{0.8}Br_{0.2})_3$ perovskite light absorbers with a one-step anti-solvent method to explore the effect of different post-treated techniques on the perovskite films. During this process, the post-treatment was performed on the conditions of thermal annealing under 100 °C for 5 min (labeled as TA) or without annealing at room temperature for 1 h (labeled as RT) with the TEACl molecule coating, as illustrated in Figure 1a. X-ray diffraction (XRD) measurements were first carried out to investigate the perovskite phases of control, TA, and RT TEACl-treated $FA_{0.75}Cs_{0.25}Pb(I_{0.8}Br_{0.2})_3$ films, as shown in Figure 1b. All three perovskite films showed characteristic XRD peaks at 14.20°, 20.10°, 28.60°, and 40.85°, corresponding to the (001), (011), (002), and (022) crystallographic planes of the perovskite films, respectively. The results also indicated that both the TA and RT TEACl treatments can improve the crystallinity of perovskite films as the XRD intensity increases after the treatments. In addition, a new peak at 5.40° was observed in the RT TEACl-treated films, corresponding to the formation of 2D perovskites of $(TEA)_2Pb(X)_4$ ($(TEA)_2(FA, Cs)_{n-1}Pb_n(X)_{3n+1}$, $n = 1$), demonstrating that a primary 2D perovskite capping layer was formed on the top surface of 3D perovskites (Figure S1, Supporting Information).^[43] For comparison, the TA TEACl-treated film exhibited several low-intensity peaks at low angles because of the permeation of TEACl salts during the annealing process

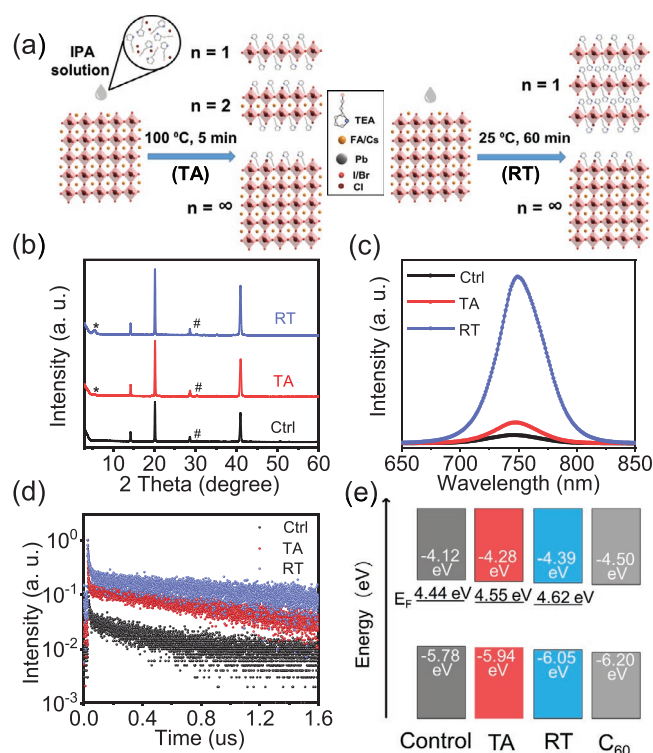


Figure 1. a) Schematic illustration of perovskite films loading TEACl post-treatment with thermal annealing at 100 °C or room temperature. b) XRD patterns, c) steady-state PL spectra, d) TRPL spectra, and e) schematic energy-level diagram of control, TA TEACl-treated, and RT TEACl-treated perovskite thin films. * and # in XRD patterns denote 2D perovskites and ITO, respectively.

and the following formation of both $n = 1$ and $n = 2$ 2D perovskite members (Figure S1b, Supporting Information).^[30,38,43] Scanning electron microscopy (SEM) images of the control and TEACl-treated perovskite films are given in Figure S2 (Supporting Information). The results reveal that all the films displayed similar morphologies whether with TEACl treatment or not. The surface characteristics of perovskite films were further investigated by X-ray photo-electron spectrometry (XPS), as shown in Figure S3 (Supporting Information). Cl 2p peak was only detected for the TEACl-treated perovskite films, suggesting that only a trace amount of chlorine was retained in the final films. Additionally, the peaks of Pb 4f from the perovskites shifted slightly to higher energies after the TEACl treatment, indicating the presence of an interaction between the TEACl molecule and Pb atoms, which could passivate the defects in perovskites.^[38]

In the following, steady-state photoluminescence (PL) and time-resolved photoluminescence (TRPL) measurements were performed to probe the carrier recombination dynamics of these perovskite films. The PL intensities of TA and RT TEACl-treated films were much stronger than that of the control film (Figure 1c). The increase in PL intensity indicated that the TEACl post-treatment effectively inhibits non-radiative recombination in the perovskite films induced by defect passivation. In particular, the RT TEACl post-treatment loading a primary $(TEA)_2Pb(X)_4$ capping layer has a much better passivation effect on the perovskite films than that of the TA TEACl

post-treatment with multi-2D perovskites. The corresponding bandgaps of these films were calculated to be 1.66 eV by the tauc plots obtained from ultraviolet-visible (UV-vis) absorption spectra, which were consistent with the bandgaps estimated from the first derivatives of the corresponding external quantum efficiency (EQE) curves (Figure S4, Supporting Information). In addition, the TRPL decay curves of these films were fitted by a bi-exponential decay equation (Figure 1d) and the corresponding results are shown in Table S1 (Supporting Information). The average lifetime (τ_{avg}) of the control film was ≈ 243.03 ns, while the τ_{avg} of TA TEACl-based and RT TEACl-based films were ≈ 930.54 and 1284.42 ns, respectively. The significantly increased τ_{avg} of perovskite films implied that the post-treatment with TEACl can effectively suppress non-radiative recombination.^[38,43] These results also indicated that the defect passivation capability of RT TEACl post-treatment was better than that of TA TEACl post-treatment tactics, which was consistent with the PL results.

The energy levels of perovskite films were investigated by Fermi level (E_F) and valence band maximum (VBM) measurements, which were directly determined by UV photoelectron spectroscopy (Figure S5, Supporting Information). Combined with the bandgap (1.66 eV) of perovskite films, Figure 1e shows a schematic diagram of the energy levels of three types of perovskite films. With the TEACl post-treatment, the E_F of the perovskite films shows a downward shift, which was also verified by Kelvin probe force microscopy measurements shown in Figure S5 (Supporting Information). Note that the

RT TEACl-treated sample has a wider energy gap between E_F and VBM in contrast with the other two samples (Figure 1e), indicating enhanced n-type characters of RT TEACl-treated perovskite films and thereby further reduced carrier recombination.^[44] Moreover, the conduction band minimum of the RT TEACl-treated perovskite film was closer to that of the electron transport layer (C_{60}), which results in more efficient charge transport at the perovskite/ C_{60} interface, as compared with the other two samples.^[40] In addition, an n/n⁺ heterojunction induced by bottom perovskites and top perovskites may also lead to an increase in a built-in electric field, which can facilitate carrier separation and suppress carrier recombination,^[45] therefore resulting in device performance enhancements.

To further examine the influence of TEACl post-treatment on solar cell performance, inverted devices were fabricated with an architecture of glass/indium tin oxide (ITO)/[2-(3,6-Dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz)/FA_{0.75}Cs_{0.25}Pb(I_{0.8}Br_{0.2})₃/C₆₀/bathocuproine (BCP)/Cu, as shown in Figure 2a. Figure 2b shows a cross-sectional SEM image of an RT TEACl-treated device. The cross-sectional SEM image of a TA TEACl-treated PSC is given in Figure S6 (Supporting Information). The results revealed that the devices with two different TEACl treatments display a similar framework. The dependence of the device performance on the TEACl concentration was also optimized. The corresponding results are summarized in Figure S7 (Supporting Information), which showed an optimal concentration of 1 mg mL⁻¹ for the TEACl post-treatment. It can be observed

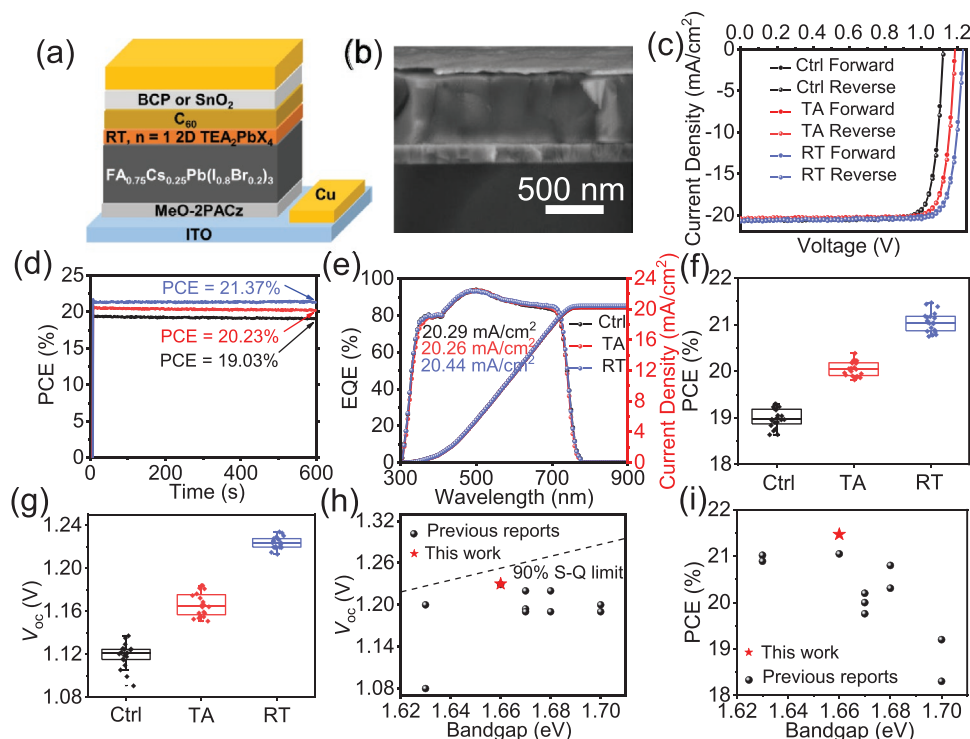


Figure 2. a) Device architecture of inverted WBG PSCs. b) A cross-sectional SEM image of a representative device with an RT TEACl post-treatment. c) J - V scans, d) steady-state power outputs, and e) EQE spectra of three types of PSCs. The bias voltages of steady-state power output measurements were 1.00, 1.04 V, and 1.08 V for the control, TA TEACl, and RT TEACl-treated devices, respectively. Statistical data of f) PCE and g) V_{oc} for three types of PSCs. Summary of h) V_{oc} and i) PCE of the reported MA-free WBG PSCs with light-absorber bandgaps of 1.63–1.70 eV (the red pentagon is the data of this work).

Table 1. Photovoltaic parameters obtained from three types of best-performing WBG PSCs.

Sample	Scan direction	V_{OC} [V]	J_{SC} [mA cm ⁻²]	FF [%]	PCE [%]
Control	Forward	1.13	20.51	83.29	19.23
	Reverse	1.13	20.49	83.37	19.29
TA	Forward	1.18	20.43	83.82	20.22
	Reverse	1.18	20.41	84.34	20.39
RT	Forward	1.23	20.75	83.23	21.16
	Reverse	1.23	20.74	84.33	21.47

that the devices with 1 mg mL⁻¹ of RT TEACl post-treatment showed the best performance with the highest PCE, V_{OC} , and comparable J_{SC} and FF in contrast with the neat devices and the devices with 0.5 mg mL⁻¹ of TEACl. Therefore, 1 mg mL⁻¹ of TEACl was applied in the following study.

Both TA and RT TEACl post-treatments can markedly improve solar cell performance. As shown in Figure 2c and Table 1, the TA TEACl-treated device showed a PCE of 20.39% (20.22%) with a V_{OC} of 1.18 V (1.18 V), a short-circuit current density (J_{SC}) of 20.41 mA cm⁻² (20.43 mA cm⁻²), and an FF of 84.34% (83.82%) under a reverse (forward) scan. After adopting a strategy of RT TEACl post-treatment on perovskite films, the optimized device even exhibited a significantly enhanced PCE of 21.47% (21.16%) and an enlarged V_{OC} of 1.23 V (1.23 V) with a reduced V_{OC} deficit of 0.43 V. By comparison, the control device only demonstrated a V_{OC} of 1.13 V (1.13 V), a J_{SC} of 20.49 mA cm⁻² (20.51 mA cm⁻²), and an FF of 83.37% (83.29%), resulting in a PCE of 19.29% (19.23%) under the same measurement conditions. The steady-state power outputs of the devices at the maximum power points (MPP) under constant AM 1.5 G illumination for 600 s are given in Figure 2d. The corresponding steady-state power outputs of the control, TA TEACl-based, and RT TEACl-based devices were measured to be 19.03%, 20.23%, and 21.37% respectively, illustrating the reliability of PCE measurements. In addition, the J_{SC} integrated from EQE spectra were $\approx$ 20.29, 20.26, and 20.44 mA cm⁻² for the control, TA TEACl-based, and RT TEACl-based devices (Figure 2e), respectively, matching well with the $J-V$ results in Figure 2c. The device performance also showed good reproducibility. As shown in Figure 2f,g, the control cells have an average PCE of $18.99 \pm 0.21\%$ with an average V_{OC} of 1.12 ± 0.01 V. And for the TA TEACl-based cells, the average PCE increased to $20.05 \pm 0.16\%$ with an average V_{OC} of 1.17 ± 0.01 V. Particularly, when loading the RT TEACl post-treatment, the averaged PCE of WBG PSCs was further improved to $21.06 \pm 0.22\%$ with an average V_{OC} of 1.22 ± 0.01 V. The results also confirmed the superiority of the RT TEACl-based tactics to the TA TEACl-based tactics. The detailed statistical results are depicted in Figure 2f,g, Figure S8, and Table S2 (Supporting Information). To estimate the effect of the 2D capping layer on the V_{OC} deficit, a detailed equilibrium theory was used to quantitatively analyze the V_{OC} deficit pathways.^[41,46] The overall voltage deficit (ΔV_{OC}) was mainly contributed by the recombination photo-voltage deficit and the non-radiative recombination voltage loss ($\Delta V_{OC, non-rad}$). The $\Delta V_{OC, SQ}$ (the recombination photo-voltage deficit at the S-Q limits) as a function of the bandgap using a

detailed balance principle was calculated to be 0.27 V. We then quantified $\Delta V_{OC, non-rad}$, which can be determined from the measured external quantum efficiency of the electroluminescence (EQE_{EL}) using the following relationship:

$$\Delta V_{OC, non-rad} = -\frac{k_b T}{q} \ln(EQE_{EL}) \quad (1)$$

As shown in Figure S9a (Supporting Information), the EQE_{EL} values at the one-sun equivalent current injection of TA and RT TEACl-treated devices were 0.03% and 0.48%, respectively, which were both higher than that of the control cell (EQE_{EL} of 0.01%). Therefore, the $\Delta V_{OC, non-rad}$ of the reference WBG PSC (control), and the devices with TA and RT TEACl post-treatments were 0.24, 0.21, and 0.14 V, respectively. According to the V_{OC} loss analysis from $\Delta V_{OC, SQ}$ and $\Delta V_{OC, non-rad}$ (Table S3, Supporting Information), the theoretical V_{OC} of the control cell and the devices with TEACl post-treatment were consistent with the V_{OC} measured from $J-V$ curves (Figure S9b, Supporting Information). In addition, the RT TEACl-based device also realized a certified and comparable PCE of 20.41% with a V_{OC} of 1.24 V and an FF of 82.97% (Figures S10, S11, Supporting Information).

Figure 2h,i and Table S4 (Supporting Information) summarize PCE and V_{OC} of the reported MA-free WBG PSCs with bandgaps of 1.63–1.70 eV (the red pentagon is the data of this work). Notably, for our champion device, the V_{OC} deficit was calculated to be merely 0.43 V, which is among the lowest values for MA-free WBG PSCs. Simultaneously, the PCE of 21.47% was also the highest report among all PSCs using MA-free WBG perovskite light absorbers with bandgaps of 1.63–1.70 eV as we know to date. Furthermore, we demonstrated that our proposed passivation technique is universal for other WBG perovskite components, such as FA_{0.8}CS_{0.2}Pb(I_{0.6}Br_{0.4})₃ with a bandgap of $\approx$ 1.77 eV, as shown in Figure S12 (Supporting Information). The PCE of 1.77 eV-bandgap FA_{0.8}CS_{0.2}Pb(I_{0.6}Br_{0.4})₃ PSC increased from 15.32% for the control cell to 17.51% for the RT TEACl-treated cell, when measured with a reverse scan (Table S5, Supporting Information). Similarly, the calculated V_{OC} loss is only 0.49 V, which holds great potential for their applications in perovskite/perovskite tandem cells.

To reveal the underneath mechanisms for the performance enhancement of WBG PSCs, space charge limited current (SCLC) tests were performed on hole-only and electron-only devices to estimate the changes in defect state density, as shown in Figure 3a and Figure S13 (Supporting Information), respectively. The used hole-only and electron-only device structures are shown in the inserts of Figure 3a and Figure S13a (Supporting Information), respectively. The defect-state density (n_{trap}) of perovskite films can be estimated according to the following formula (2).^[47]

$$n_{trap} = \frac{2\epsilon\epsilon_0 V_{TFL}}{qd^2} \quad (2)$$

where n_{trap} is the trap density, ϵ is the relative dielectric constant ($\epsilon \approx 24.4$),^[48] ϵ_0 is the permittivity of vacuum ($\epsilon_0 \approx 8.854 \times 10^{-12}$ F m⁻¹), q is the elementary charge, and d is the thickness of the perovskite layer ($d \approx 550$ nm). Based

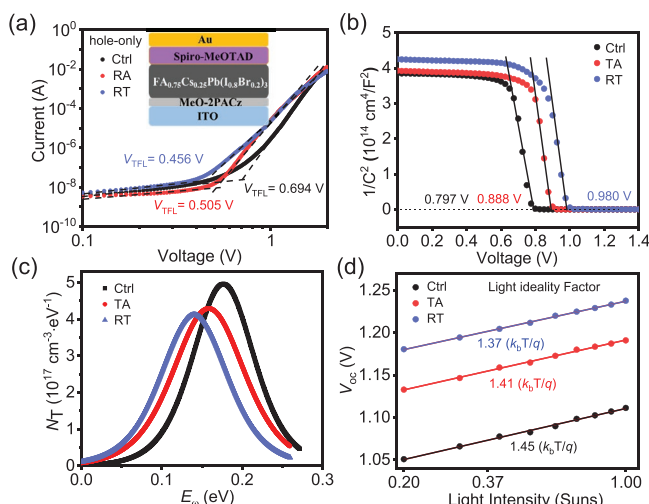


Figure 3. a) SCLC curves of control, TA TEACl-treated, and RT TEACl-treated hole-only devices. b) Mott-Schottky analysis of control, TA TEACl-treated, and RT TEACl-treated PSCs at 1 kHz in the dark. c) Trap state densities (N_T) of these three devices measured at 300 K. d) Light-intensity-dependent V_{OC} comparison of these three PSCs.

on dark current-voltage curves of the hole-only devices, we could estimate that the trap-filled limit-voltages (V_{TFL}) are 0.694, 0.505, and 0.456 V for the control, TA TEACl-treated, and the RT TEACl-treated WBG perovskite films, respectively (Figure 3a). According to Equation (2), the corresponding n_{trap} of the three samples were $\approx 6.20 \times 10^{15}$, 4.50×10^{15} , and $4.07 \times 10^{15} \text{ cm}^{-3}$. The electron-trap densities obtained from the three samples also showed the same trend, as shown in Figure S13 (Supporting Information). These results implied that the RT TEACl post-treatment was more beneficial to passivating both hole and electron traps in comparison with TA TEACl post-treatment tactics, agreeing with the prolonged carrier lifetime of the film with RT TEACl post-treatment shown in Figure 1d.

Capacitance–voltage (C – V) measurements were implemented to estimate the built-in potential (V_{bi}) and depletion width (W) of the devices according to a depletion approximation equation (Mott–Schottky) as follows.^[49]

$$\frac{A^2}{C^2} = \frac{2(V_{bi} - V)}{q\epsilon\epsilon_0 N} \quad (3)$$

$$W = \sqrt{\frac{2\epsilon\epsilon_0 V_{bi}}{qN}} \quad (4)$$

where A , C , V , and N are the active area, capacitance, applied bias voltage, and doping profile, respectively. The V_{bi} can be captured from the intersection on the voltage axis by linear region fitting the C^{-2} – V curves as shown in Figure 3b. The V_{bi} of the reference WBG PSC (control) and the devices with TA and RT TEACl post-treatments were 0.797, 0.888, and 0.980 V, and the corresponding W values were 94, 122, and 130 nm, respectively. Obviously, the improvement in V_{bi} springs from the decrease in non-radiation recombination after surface passivation, which results in a small V_{OC} deficit.

In addition, to further reveal the passivation effect of the two tactics more directly and clearly, temperature-dependent admittance spectroscopy (AS) measurements were carried out on the devices without or with TEACl treatment. AS is an effective means for evaluating both the energy level of trap states and the distribution of trap state densities.^[50–52] E_a is the difference between the trap state location and the valence band maximum.^[52] The relation between E_a and characteristic transition angular frequency (ω_0) is as follows:^[49]

$$\omega_0 = \beta T^2 \ln \left(-\frac{E_a}{k_b T} \right) \quad (5)$$

where β is a temperature-independent parameter, T is the temperature and k_b is Boltzmann's constant. The ω_0 can be determined by the derivative of the capacitance-frequency spectrum. The distribution of trap state density (N_T) and the energy level of trap states can be derived from the following Equations (5) and (6),^[49] respectively.

$$E_\omega = k_b T \ln \left(\frac{\beta T^2}{\omega} \right) \quad (6)$$

$$N_T(E_\omega) = -\frac{V_{bi}}{qW} \frac{dC}{d\omega} \frac{\omega}{k_b T} \quad (7)$$

The capacitance spectra at various temperatures of PSCs without and with TEACl treatment are shown in Figure S14a–c (Supporting Information). Moreover, the corresponding Arrhenius plots of the characteristic transition frequencies determined from the corresponding derivative capacitance spectra are presented in Figure S14d–i (Supporting Information).

The calculated E_a of the pristine PSC and the devices with TA and RT TEACl post-treatment tactics were 0.175, 0.158, and 0.140 eV, respectively. It showed that the E_a shifted from 0.175 to 0.140 eV above the valence band maximum after TEACl treatment and the trap state density was also slightly diminished. Then, combined with the AS results, the N_T under room temperature in these devices can be finally calculated, and the results are illustrated in Figure 3c. Note that the E_a of the perovskite with RT TEACl post-treatment shifts from 0.158 to 0.140 eV above the valence band maximum and the integrated defect density correspondingly decreases from 4.91×10^{16} to $4.52 \times 10^{16} \text{ cm}^{-3}$, as compared with those of the one with TA TEACl post-treatment. These results verified the excellent trap-state passivation capability of the RT TEACl treatment technique compared with that of the TA TEACl one.

To further investigate the charge carrier recombination losses in devices, light intensity-dependent J – V characteristics were carried out. The V_{OC} versus the semi-natural logarithm of light intensity presents a linear relationship, as shown in Figure 3d. The plot slopes of the pristine PSC and the devices with TA and RT TEACl post-treatment were 1.45 $k_b T/q$, 1.41 $k_b T/q$, and 1.37 $k_b T/q$, respectively. The RT TEACl-based device with a decreased slope suggests the reduced trap-assisted bimolecular recombination at the perovskite/ C_{60} interface,^[40,53] thus leading to improved V_{OC} and FF of WBG PSCs. The above results confirmed that the RT TEACl was better than the TA TEACl

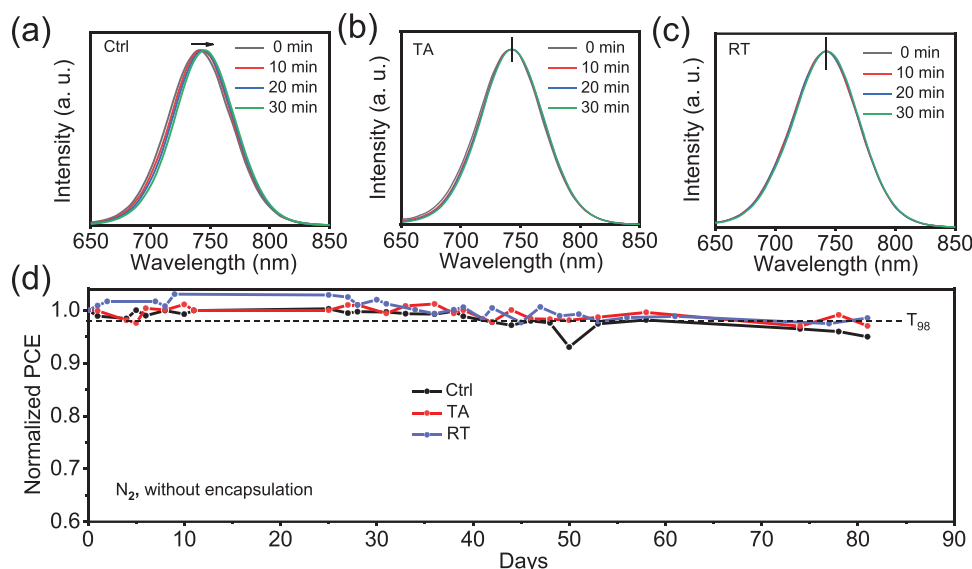


Figure 4. a–c) Normalized PL spectra of control, TA TEACl-treated, and RT TEACl-treated WBG perovskite films under constant illumination of a 481 nm laser. d) PCEs of the unencapsulated devices as a function of storage time in an N₂-filled glovebox.

post-treatment technique. Therefore, the RT TEACl post-treatment technique can further reduce V_{OC} loss and improve the performance of WBG PSCs. These results were in good agreement with the other results described above.

Subsequently, the stability of the WBG perovskite films and devices was investigated. Figure 4a–c shows the change of PL peak positions of the perovskite films under constant illumination of a 481 nm laser. After 30 min of constant illumination, the TA and RT TEACl-treated perovskite films showed stable PL peak positions, while the peak position of the control sample exhibited a redshift, indicating photo-induced perovskite phase separation. Barker et al. demonstrated that the halogen vacancies on the perovskite surface were responsible for halide migration, which facilitated the formation of iodide-rich and bromine-rich domains on the WBGs perovskite surface.^[25] In addition, an electrostatic potential difference might be established between the electronic traps and the hole traps at the perovskite surface, which would act as the driving force for halide segregation.^[54,55] In our case, TEACl can supply chlorine to fill the halide vacancies and TEA to coordinate the uncoordinated metal cations (Figure S3, Supporting Information), which could reduce ion migration channels and local electric fields, thereby inhibiting perovskite phase separation. Figure 4d displays the storage stability test of the unencapsulated devices in an N₂-filled glovebox. It was found that the PCEs of both the RT and TA TEACl-treated devices sustained over 98% of their pristine efficiencies after 81 days. However, the control device kept $\approx 98\%$ of its initial efficiency after 40 days and only kept $\approx 95\%$ of its pristine efficiency after 81 days. These results indicated that the post-treatment tactics can improve the stability of perovskite solar cells.

Encouraged by the improved performance of WBG PSCs, we then set out the study on tandem devices. We fabricated 4T all-perovskite TSCs assembling semi-transparent WBG PSCs as a top cell and NBG PSCs as a bottom cell without an antire-

flection layer. The device structures of semi-transparent WBG PSCs, NBG PSCs, and 4T tandem cells are given in Figure 5a. Note that the utilization of an atomic layer deposited SnO₂ (20 nm) instead of BCP can weaken the damage to the WBG perovskite layer in the process of sputtering an ITO transparent top electrode. Remarkably, the semi-transparent WBG PSC delivers a high PCE of 19.11% (19.13%), a V_{OC} of 1.18 V (1.18 V), a J_{SC} of 19.30 mA cm⁻² (19.26 mA cm⁻²), and an FF of 83.63% (83.94%) with a reverse (forward) scan, as seen in Figure 5b and Table S6 (Supporting Information). The transmittance spectrum of the semi-transparent PSC is given in Figure S15 (Supporting Information). The integrated J_{SC} of the semi-transparent WBG device calculated from the EQE spectrum is 19.01 mA cm⁻² (Figure 5c), matching well with the J_{SC} obtained from the J - V curves (Figure 5b). Figure 5d shows the stable output of the semi-transparent WBG device, which achieved a stabilized PCE of 19.16% under constant illumination at the MPP for 65 s. As we know to date, the PCE of the semi-transparent WBG device reported in this work is also among the highest values for inverted semi-transparent MA-free WBG PSCs. After light-filtered by the semi-transparent WBG perovskite top cells, the PCE of the NBG bottom cell employing a 1.25 eV-bandgap mixed Pb-Sn perovskite absorber reaches 7.53% with a J_{SC} of 11.66 mA cm⁻², which was also consistent with the EQE data (Figure 5b,c, Table S6, Supporting Information). Note that silicone oil was applied in the air gap between the WBG PSCs and the NBG PSCs to minimize optical loss. J - V curves of the original NBG PSC and its photovoltaic parameters are given in Figure S16 and Table S6 (Supporting Information). Figure 5d also shows that the stable output of the filtered NGB PSC realizes a stabilized value of 7.39%. Finally, a 4T TSC assembled with two subcells reaches a stable output of 26.55% (Figure 5d). These superior results indicated that the RT TEACl post-treatment technique can facilitate the achievement of high-efficiency perovskite/perovskite tandem solar cells.

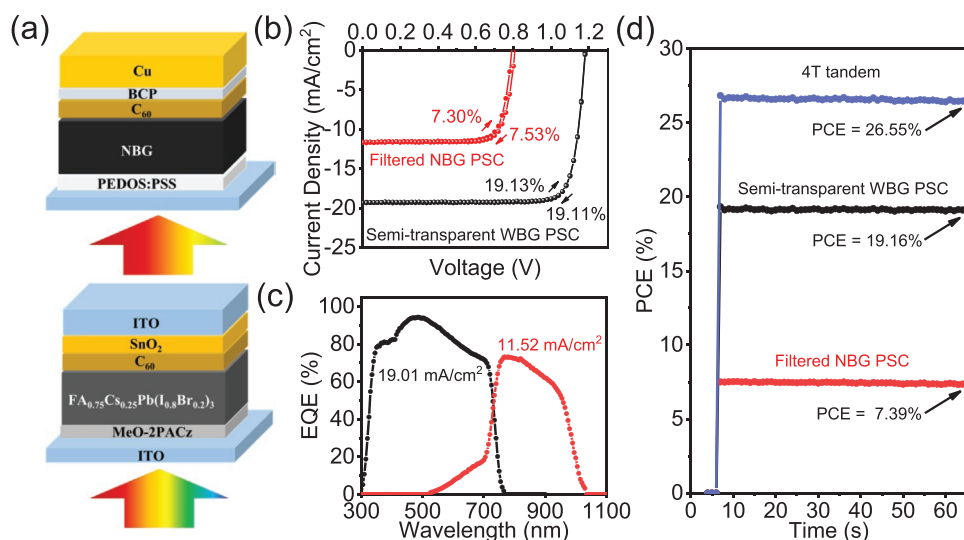


Figure 5. a) Device architecture of perovskite/perovskite 4T tandem solar cells. Silicone oil was applied in the air gap between the WBG PSCs and the NGB PSCs. b) J - V curves and c) EQE spectra of a single-junction NGB PSC, a single-junction semi-transparent WBG PSC, and their tandem cell. d) Steady-state power outputs measured at bias voltages of 1.04 and 0.68 V for a semi-transparent WBG PSC and a filtered NGB PSC, respectively.

3. Conclusion

In summary, we demonstrated high-efficiency MA-free WBG PSCs by using an RT post-treatment tactic with a TEACl molecule to form a 2D $n = 1$ TEA₂PbX₄ top layer on perovskites. This method can passivate the defect of perovskite films more effectively, prolong carrier lifetimes, and suppress photo-induced perovskite phase segregation, as compared with the neat one and the one with TA post-treatment tactics. As a result, the devices achieved much-improved PCE, V_{OC} , and FF, reaching the highest PCE of over 21% with a remarkable V_{OC} of 1.23 V and FF of 84.33%. Furthermore, a PCE of 19.11% was obtained from the optimized semi-transparent device. By coupling the semi-transparent PSC with a 1.25 eV-bandgap Pb-Sn perovskite bottom cell, a 4T perovskite/perovskite TSC successfully delivered a decent PCE of 26.64%. Our method would pave the way to the fabrication of high-efficiency WBG PSCs as well as the development of tandem solar cells.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords

open-circuit voltage deficit, perovskite tandem solar cells, phase segregation, wide-bandgap perovskites

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4. Experimental Section

Experimental details are provided in the Supporting Information.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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